Quan Liu, Ph.D.

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Summary

LLM/Agent Research Scientist & Tech Lead building production-grade multi-agent systems at scale.
Led development of AIRefinery (Accenture's enterprise agent platform), adopted across 30+ enterprise use cases, including Fortune 500 companies.
Expert in agent orchestration, tool grounding (MCP-style), RAG systems, and post-training alignment (SFT, PPO, GRPO).

Education

Vanderbilt University
Ph.D., Computer Science *2024*

Huazhong University of Science and Technology (HUST)
B.S., Electrical Engineering *2018*

Experience

Accenture - Center for Advanced AI **Mountain View, CA**
Sr. Research Scientist / Tech Lead *2024–Present*

- **Agent Systems:** Architected and deployed AIRefinery, a production multi-agent platform enabling agent-as-a-service across enterprise workflows, supporting 1K+ concurrent users.
- **Platform:** Contributed to AIRefinery adoption across enterprise use cases and internal developer ecosystems, featured in major AI industry events (e.g., NVIDIA GTC)
- **Agent Orchestration:** Led development of core agent framework (Distiller), including orchestrator, planning agents, and dynamic tool chaining
- **Memory Systems:** Designed multi-level agent memory and short/long memory dynamic indexing (chat, agent-specific, cross-agent) for long-horizon reasoning and coordination
- **Tool Use / MCP-style:** Built schema-constrained tool execution pipeline (tool grounding, validation, execution tracing), ensuring reliable agent behavior under enterprise constraints
- **RAG / Retrieval:** Designed hybrid retrieval system (E5 + BM25 + FAISS) over 50M+ documents, improving precision by 12% and reducing hallucination by 18%
- **Evaluation:** Developed automated evaluation pipeline (LLM-as-judge + task-based metrics) for scalable agent benchmarking and quality assessment
- **Post-Training:** Designed end-to-end alignment pipeline including:
  Data - Curated failure-driven datasets to improve edge-case handling and reduce model errors
  LLM - Trained and aligned LLMs (Llama-3, Mistral) using SFT, PPO, GRPO on 30B+ tokens, improving reasoning performance
  Agent - Trained LLMs for multi-agent coordination and tool planning, improving long-horizon task completion
- **Leadership:** Led a team of 6 engineers, driving system design, code reviews, and delivery of core agent infrastructure across multiple client deployments

Merck - Image Data Analytics Team **West Point, PA**
Research Intern *2023*

- Designed multimodal fusion pipeline integrating imaging and structured data for robust prediction
- Reduced inference cost by 40% via anomaly-guided patch selection and efficient visual encoding

Alibaba - DAMO **New York, NY**
Research Intern *2022*

- Built multimodal learning framework (CLIP-based) combining vision-language representations for mutation classification (+9 F1 score)
- Designed missing-modality retrieval and alignment methods for real-world multimodal deployment

Selected Publications

MCP-Bench: Benchmarking Tool-Using LLM Agents with Real-World Tasks — ICLR 2026, NeurIPS 2025
DRAGON: Guard LLM Unlearning via Negative Detection and Reasoning — ICLR 2026
PromptBridge: Cross-Model Prompt Transfer for Large Language Models — arXiv
LLM Unlearning via Loss Adjustment with Only Forget Data — ICLR 2025

Skills

LLM & Agents: Multi-agent systems, planning, tool use, memory, RAG, MCP-style integration
Alignment: SFT, DPO, PPO, GRPO, RLHF